effects. Although the action you perform is the same, the effect is different because it's based on the variation in the data. You'll see data-binding first at the end of this chapter, and in much more detail throughout this book.

### 1.2.4 Web page elements can now be divs, countries, and flowcharts

We've grown accustomed to thinking of web pages as consisting of text elements with containers for pictures, videos, or embedded applications. But as you grow more familiar with D3, you'll begin to recognize that every element on the page can be treated with the same high-level abstractions. The most basic element on a web page, a `<div>` that represents a rectangle into which you can drop paragraphs, lists, and tables, can be selected and modified in the same way you can select and modify a country on a web map, or individual circles and lines that make up a complex data visualization.

To be able to select items on a web page, you have to ensure that they're built in a manner that makes them a part of the traditional structure of a web page. You can't select items in a Java applet, or in a Flash runtime, nor can you select the labels on an embedded Google map, but if you create these elements so that they exist as elements in your web page, then you give yourself tremendous flexibility. To get a taste of this, look at chapter 7, where we'll build robust mapping applications in D3, and we'll use the `d3.select()` syntax to update the appearance of a mapping application in the same manner as it's being used here and elsewhere to create and move circles or `<div>` elements.

## 1.3 Using HTML5

We've come a long way from the days when animated GIFs and frames were the pinnacle of dynamic content on the web. In figure 1.10, you can see why GIFs never caught on for robust data visualization on the web. GIFs, like the infoviz libraries designed to use VML, are still necessary for earlier browsers, but D3 is designed for modern browsers that don't need the helper libraries necessary for backward compatibility. D3 development isn't for everyone, but if your audience can be assumed to have access to a modern web browser, D3 also brings a significant reduction in the cost necessary not only to code for older browsers but also to learn and keep updated on the various libraries that support backward compatibility with those older browsers.

A modern browser typically can not only display SVG graphics and obey CSS3 rules, but also has great performance. Along with Cascading Style Sheets (CSS) and Scalable Vector Graphics (SVG), we can break down HTML5 into the DOM and JavaScript. The following sections treat each of them and include code you can run to see how D3 uses their functionality to create interactive and dynamic web content.

### 1.3.1 The DOM

A web page is structured according to the DOM. You need a passing familiarity with the DOM to do web development, so we'll take a quick look at DOM elements and structure in a simple web page in your browser and touch on the basics of the DOM. To get started, you'll need a web server that you can access from the computer that